

ground. If you need to change output polarity, Cuk topology is a good option. Transformers provide isolation, and topologies that incorporate it in the design include flyback, clamp forward, push-pull, half-bridge or full-bridge.

Types of power supplies for electronic devices

Most electronic devices work on regulated DC, independent of input supply (AC or DC). Depending on the power supply (AC or DC), all such devices have a different setup to convert it into regulated DC supply. Variable AC power supply is important to evaluate the equipment when it is exposed to over- or under-voltage conditions.

Different types of DC power supplies are as follows:

Uninterruptible power supply. This is a backup power source in case of power failure or fluctuations.

Constant voltage supply. This is adjustable and usually supplied with a meter to show the voltage the supply is set to.

Constant current supply. When in constant current mode, the power supply maintains the set current regardless of changes in load resistance.

Multiple output supply. This has more than one DC output, and is useful and cost-effective for systems that require multiple voltages.

Programmable supply. This is often used as part of a computer-operated system for testing or production.

Multi-range supply. It operates with fixed voltage and current ratings, such as 30V, 3A.

When there is a need for special characteristics or unusual form factors, custom designs are done particularly for harsh environments like in military or avionics.

Specifications to look for

Power supplies have different specifications for different applications. While choosing a power supply, each specification must be considered. It is necessary to understand everything from voltage and current ratings to ripple, load regulation, input voltage regulation and the like. A few important parameters to be taken care of

TABLE II LINEAR POWER SUPPLY VERSUS SWITCH-MODE POWER SUPPLY		
Description	Linear power supply	Switch-mode power supply (SMPS)
Advantages	• Low noise/ripple and EMC disturbances • Good line and load regulation • Fast transient response • Can produce very low current output	• High efficiency 75 per cent average • More affordable for higher power • Lighter
Disadvantages	• Low efficiency (30 to 40 per cent average) • Large and heavy • Larger heatsinks • More expensive for higher power	• More EMI • Much slower transient response

are given below.

- Noise and ripple, which is a small variation in voltage when AC is transformed into DC

- Low dropout to reduce power losses

- Fast dynamic response

- Line regulation, which is the variation in output voltage with change in input or line voltage

- Load regulation, which is the variation in output voltage for a change in load

- Over-current, which can be controlled by internal protection circuit

- Over-temperature, which can be controlled using a temperature sensor and associated circuit

- Over-voltage, which can be controlled by the supply with over-voltage protection circuit

- Electromagnetic compatibility, which uses a design technique to minimise electromagnetic interference (EMI)

- Isolation. An isolated power supply employs a transformer to eliminate the DC path between input and output, whereas a non-isolated one follows a DC path between input and output of the supply

Peeyush Rustagi, assistant manager, ANTPL says, "It is better to have specifications like input voltage range (AC or DC), output voltage (DC or AC) and tolerances, output current requirement, ripple maximum, estimated total power requirement, efficiency requirement, EMI considerations, electrical isolation between input and output requirement, temperature range of operation, peak and average output current, transient exposure and response needs, load- and line-regulation requirement, switching frequency and need for power factor correction. Efficiency is one of the most important

power supply characteristics to judge system performance."

Nishu Kumar Rauniyar, hardware designer, SGS Weather & Environmental Systems, says, "Some power delivery architecture needs to be followed while supplying power for different ICs. Special care is required for high-frequency components, as RF noise can destroy the whole PSU. To work as intended, a proper grounding system is necessary. Special attention needs to be paid to layout design, as it will decide overall system performance."

Preferred applications

A PSU is used in all electronic equipment that use DC voltage as input, such as radio transmitters, receivers, computers and so on. Switching power supplies are often used in applications where battery life and temperatures are important, such as electrolysis, waste treatment, fuel cells, DC motors, slot cars, aviation and marine, R&D, manufacturing and testing equipment, battery charging for lithium-ion batteries and electroplating.

Linear regulators provide noise-free output voltage but have poor efficiency and are large in size. This makes them suitable for devices that require high frequency and low noise, such as signal processors, control circuits, automated and laboratory test equipment, low-noise amplifiers, data-acquisition sensors and circuits, among others.

For selecting an appropriate power supply for an electronic device, it is important to first understand the requirement of the setup. Explore the different types of power supplies available in the market. Check for the required specification and the preferred application. **EFY**